

Beyond the Complaint Count: How Toronto 311 Service Requests Have Changed Over Time

Milestone 3 – Initial Coding & Results

Kezia Mathew

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Supervisor: Dr. Tamer Abdou

Toronto Metropolitan University

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Project Overview

This project analyzes Toronto 311 customer-initiated service request data to examine how reported municipal service demand has changed over time. The analysis compares two four-year periods: 2010–2013, representing the earliest full-year period available in the dataset, and 2022–2025, representing the most recent complete full-year period.

The project is written from the position of a municipal data analyst. This project supports how municipal staff can use historical and recent 311 request patterns to understand reported service demand while avoiding unsupported claims about actual community need. This distinction is important because 311 request volume may reflect not only service issues, but also changes in reporting behaviour, public awareness, access to 311, service categories, and data publication practices.

The analysis will focus on service request categories, city divisions, monthly and seasonal patterns, and broad reporting profile changes. Ward-level comparisons will be treated cautiously because ward structures changed between the early and recent periods and should be noted.

Changes Made

Based on the feedback provided, below is the outline of changes made:

- **Framework:** the project is now framed as a municipal data analyst project supporting city of Toronto service planning and operational monitoring.
- **Revised time-period** the original focus was 2022–2025. After further data exploration, I added 2010–2013 as an early comparison period so the project can examine how reported 311 request patterns have changed over time, to potentially generate stronger analysis.
- **Refined research questions:** The research questions were revised to have exactly three non-overlapping questions, each with a clear purpose, method, expected output, and stakeholder use.
- **Expanded dataset and audit plan:** The report will document file format, date coverage, key fields, geography, update frequency, coverage limits, and comparability issues between the early and recent periods.
- **Added baselines and comparison points:** The analysis will compare early and recent periods using frequency counts, percentage shares, cumulative shares, seasonal variation, and data quality checks.
- **Strengthened reproducibility and GenAI controls:** An initial EDA report and supporting notebook will be uploaded to GitHub, and the GenAI statement will include risks such as incorrect code, fake citations, and unsupported interpretations.

Decision Context: Who Benefits and Who is it for?

This project is written from the perspective of a municipal data analyst supporting Toronto's service planning and operational decision-making. The main users of this analysis would be operations staff, public services workers, policy analysts, and city council members.

The analysis supports those who need to understand how consumer reported 311 service request patterns have changed over time. The project helps to identify whether the most common service request categories, city divisions, and seasonal patterns have shifted by comparing the earliest (2010-2013) and most recent (2022-2025) full-year periods. The analysis can understand where 311 requests are showing up repeatedly, which services may be getting more reports, and where staff may need to look more closely before making decisions. For example, if snow-related requests always rise in certain months, or if waste requests make up a larger share in the recent period, that does not automatically mean the city should move resources right away. It means the pattern is worth reviewing.

This project does not decide where the city should put its staff, funding, or services. Instead, it helps decision-makers interpret the open 311 dataset more carefully by showing what the data can reveal, what it cannot prove, and where caution is needed because of changes in categories, divisions, wards, data coverage, or reporting behaviour.

Revised Research Questions

The research questions were revised to create exactly three non-overlapping questions that support the overall project goal. Each question focuses on a different part of the analysis: demand concentration, seasonal change, and responsible comparison across time periods.

Overall Research Question:

How have Toronto 311 reported service request patterns changed between 2010–2013 and 2022–2025, and what limitations should be considered when using these patterns for municipal service planning?

RQ1: How has reported 311 demand concentration by service request category and city division changed between 2010–2013 and 2022–2025?

This question examines whether the same service request categories and city divisions dominate reported demand in both periods, or whether the composition of 311 requests has changed over time. The analysis will use request counts, percentage share, cumulative share, and ranking comparisons.

RQ2: Which high-volume 311 service request categories show different monthly or seasonal patterns between 2010–2013 and 2022–2025?

This question examines whether the timing of reported demand has changed across the early and recent periods. The analysis will compare monthly and seasonal request patterns for high-volume categories to identify recurring or changing seasonal trends.

RQ3: Which differences between the early and recent periods can be compared responsibly, and which require caution because of changes in categories, divisions, wards, data coverage, or reporting practices?

This question focuses on interpretation and data limitations. It examines which observed changes are useful for municipal planning and which should be interpreted cautiously because the dataset structure, ward boundaries, category labels, or reporting practices may have changed over time.

Research Question Alignment Table:

Research Question	Required Fields	Planned Method	Expected Output	Stakeholder Use
RQ1: How has reported 311 demand concentration by service request category and city division changed between 2010–2013 and 2022–2025?	Creation Date, Service Request Type, Division, Section, Status, request count	Combine all CSV files into one dataset. Create a period variable to separate 2010–2013 and 2022–2025. Group requests by Service Request Type and Division for each period. Also, need to fix how the creation date is formatted, it's showing as an object which isn't correct. Calculate which service types and city divisions make up the most of the 311 requests by using the methods of ranking, percentage share, calculating counts, cumulative percentage share, and compare patterns present in both periods.	Top category tables for each period, top division tables, percentage share comparison, cumulative share/Pareto-style chart, ranking comparison table.	Helps operations staff and service managers understand whether reported workload has shifted across service categories and city divisions over time.
RQ2: Which high-volume 311 service request categories show different monthly or seasonal patterns between 2010–2013 and 2022–2025?	Creation Date, Service Request Type, Division, year, month, season (code by month: Spring (March–May), Summer (June–August), Fall (September–November), and Winter (December–February)), request count	Convert Creation Date into datetime format. Create year, month, and season fields. Convert months into season: Spring, Summer, Fall, and Winter Use the high-volume categories identified in RQ1. Group requests by month, season, category, and period. Compare seasonal peaks, monthly trends, and variation across the early and recent periods.	Monthly trend charts, seasonal comparison charts, category-by-month heatmap, table of peak months/seasons, seasonal variation summary.	Helps service managers and city divisions identify recurring or changing seasonal pressures in reported 311 demands.
RQ3: Which differences between the early and recent periods can be compared responsibly, and which require caution because of changes in categories, divisions, wards, data coverage, or reporting practices?	Service Request Type, Division, Ward, Creation Date, Status, missing values, unique category counts, unique division counts, unique ward counts, data coverage notes	Conduct a data audit comparing the structure of the early and recent datasets. Check field consistency, missing values, duplicate records, category changes, division changes, ward structure differences, and date coverage. Identify which comparisons are reliable, limited, or not appropriate.	Data comparability table, limitation matrix, missingness summary, field consistency summary, responsible interpretation notes.	Helps policy analysts, councillors' offices, and municipal decision-makers avoid misleading conclusions when comparing 311 request patterns across time.

Methodological Rational and Literature Review

The 311 Toronto service officially launched on September 24, 2009, with the purpose of giving residents and businesses one easy way to request information or submit a non-emergency service request to the city (Deloitte, 2013, p. 6). Since then, 311 has become an important part of how Toronto communicates with the public and organizes service delivery (Stowers, 2022, p. 14). Instead of residents having to figure out which City division or department to contact, 311 provides a centralized system where service requests can be submitted, tracked, and directed to the appropriate area. The main goal of this project is to examine how customer-initiated 311 service demand has changed over time by comparing the earliest complete period available for this project, 2010–2013, with the most recent complete period, 2022–2025. The dataset is available through the City of Toronto Open Data Portal in CSV format. Each row represents one customer-initiated service request, which makes the dataset useful for grouping and comparing records by time-period, service request type, city division, month, and season.

The literature supports using 311 data to study reported municipal service demand. Stowers (2022) uses real 311 service request data from open data portals across 29 cities and treats each service request as the unit of analysis. This is similar to my project because each row in the Toronto 311 dataset will also be treated as one reported service request. Stowers also uses descriptive statistics, service categories, and visualizations to compare 311 systems. This supports my use of counts, percentage shares, rankings, tables, and charts to identify which service areas appear most often in the data. Stowers (2022, p. 17) also shows why it is important to examine the types of service requests, not only the total number of requests. In that study, common categories included garbage and recycling, code enforcement, parking, bulk item pickup, and abandon vehicles. This matters for my project because total 311 request volume alone does not explain what kinds of issues residents are reporting. To understand how reported demand has changed over time, I need to compare both the number of requests and the types of requests between 2010–2013 and 2022–2025. Research on 311 systems also shows that request volume can be influenced by resident behaviour, not only by the number of service problems in an area. Stowers (2022, p. 16) summarizes studies showing that residents may be more likely to report issues when they feel a sense of concern or “ownership” over their neighbourhood. Other research also suggests that 311 use can be shaped by technology use, satisfaction with city services, neighbourhood conditions, and the characteristics of the people living in an area. This is important for my project because 311 request counts should be interpreted as reported demand, not as a complete measure of actual municipal need.

Wang et al. (2017, pp. 4–5) supports looking at the structure of 311 service categories to identify patterns in the data. In their study, they created a 311 “pattern signature” by calculating the relative frequency of each service request type within an area. This means they did not only look at the total number of 311 requests but also examined what share of requests belonged to each category. This is useful for my project because I am also comparing the composition of 311 requests, not just the total volume. By using counts, percentage shares, rankings, and category comparisons, I can identify whether the types of reported 311 demands changed between 2010–2013 and 2022–2025. Wang et al. (2017, p. 5) used these category-based patterns to group areas with similar 311 request profiles through clustering. While my project does not rely on clustering

as the main method, the same logic supports my planned analysis. The categories in the 311 datasets can help reveal patterns in how residents report municipal service issues. For my project, this means service request types can be used to compare early and recent reporting profiles and to identify which categories appear more common, less common, newly added, or difficult to compare over time. The Toronto Open Data guest post comments on how the analysis is affected by changing request categories, ward boundary changes, missing fields, and partial dataset coverage (Lal, 2026). For this reason, my project treats the findings as patterns in the available open 311 records, not as a complete measure of all 311 demand in Toronto.

White and Trump (2018) provide another important caution. They explain that 311 requests should not automatically be treated as a measure of civic participation or actual neighbourhood need. Request volume can be affected by many factors, including whether residents know about 311, whether they trust the service, whether they have time to report issues, and whether they face language or access barriers (White and Trump, 2018, p. 15-17). This supports one of the main limitations of my project. A high number of requests in one area does not automatically mean that area has worse service conditions. A low number of requests also does not mean there are fewer problems. It may simply mean fewer people are reporting issues through 311. This is especially important for any geographic analysis in my project. Since the dataset does not include demographic information about who submitted each request, I cannot make strong claims about resident behaviour, inequality, or actual service need. I can describe where reported requests appear in the data, but I need to be careful not to overstate what those patterns mean.

Based on the literature, my project starts with a simple descriptive baseline method and then adds more careful comparison steps. First, I will combine the yearly CSV files and label each record by year and period. Second, I will profile the dataset by checking column structure, missing values, duplicates, date coverage, service request types, division labels, ward values, and status values. Third, I will compare service request categories and divisions using counts, percentage shares, rankings, and cumulative shares. Fourth, I will create date-based fields such as year, month, and season to study time-based patterns. Finally, I will include a limitation and comparability review to explain which findings can be compared clearly and which ones need caution. Overall, the literature supports the main design of my project. 311 data is useful for studying reported municipal service demand, but it is not a perfect measure of actual need. For that reason, I will combine descriptive statistics, category comparisons, division summaries, seasonal analysis, data profiling, and limitation analysis rather than using just the plain data to determine demand and patterns.

Baseline Method and Gap Analysis

The baseline method for this project is simple frequency analysis. This means counting 311 service requests by service request type, city division, year, month, season, and comparison period. The baseline outputs would include frequency tables, top category rankings, top division rankings, bar charts, and basic percentage share tables.

This baseline is useful because it gives a clear first look at the dataset. It can show which request categories appear most often, which city divisions receive the most reported requests, and whether the total number of requests differs between 2010–2013 and 2022–2025. It is also easy

to reproduce and easy for non-technical stakeholders to understand. For this reason, basic descriptive analysis is an appropriate starting point for this project.

However, the baseline method is limited because it mainly answers, “what appears most often?” It does not fully explain whether the early and recent periods are comparable, whether categories changed over time, or whether certain request types follow seasonal patterns. A simple top 10 ranking may show that one category has a higher count in the recent period, but it does not explain whether that change reflects a real shift in reported demand or a change in how the city labelled, grouped, or published the request type. The planned analysis improves on the baseline by adding more context to the counts. Instead of only reporting totals, the project will calculate percentage shares, rankings, and cumulative shares so that the early and recent periods can be compared more fairly. Monthly and seasonal variables will also be created from the Creation Date field to identify recurring patterns across the year. Finally, a comparability check will be used to flag issues such as changed category names, division labels, ward values, missing fields, and other differences that may affect interpretation.

The gap between the baseline and the planned method is important because this project is not just a “top 10 complaints” analysis. The goal is to understand how reported 311 demand patterns changed over time and how carefully those changes should be interpreted. The baseline shows the basic distribution of requests, while the planned method adds the context needed to compare the two periods more responsibly.

Planned Analysis Method

This project will use Python as the main tool for cleaning and analyzing the Toronto 311 data. SweetViz may be used to create an automated data profile, and Power BI may be used later to make simple dashboard-style visuals.

I will first combine the yearly CSV files for 2010–2013 and 2022–2025 into one dataset. I will add new columns for year, month, season, and comparison period so the early period and recent period can be compared clearly. I will also check the dataset for missing values, duplicate records, data types, date coverage, and unique values in important fields.

After, I will clean the main fields used in the analysis. The Creation Date column will be converted into a proper date format. Fields such as Service Request Type, Division, Section, Status, Ward, and the first three characters of the postal code will be checked for missing values, formatting issues, spelling differences, and category changes across years. The main analysis will use descriptive statistics (mean, median mode), frequency counts, percentage shares, rankings, and cumulative shares. Most of the dataset is categorical, so I will group them. This includes total requests, mean monthly requests, median monthly requests, minimum and maximum monthly requests, range, and standard deviation where applicable.

I will also compare patterns by year, month, and season to see whether certain service request types are more common at specific times of the year. Cross-tabulations and heatmaps will be used to compare relationships such as period by service request type, period by division, and season by service request type. Bar charts, line charts, stacked bar charts, and cumulative share charts will be used to show the main findings.

Before interpreting the results, I will complete a comparability audit. This will check whether service request labels, division names, ward values, missing values, and date coverage changed between the early and recent periods. If time allows, I may include one simple hypothesis test to compare whether the percentage share of a major request type changed between the two periods.

I will not use methods such as NLP, Naive Bayes, regression, cross-validation, or normal probability modeling as main methods because the current dataset is mostly structured, categorical, and time-based. These methods would make more sense as future extensions if the dataset included full text descriptions, outcome labels, or a forecasting goal.

The final outputs will include a cleaned dataset, SweetViz profile if completed, summary tables, ranking tables, percentage share comparisons, seasonal charts, heatmaps, optional Power BI visuals, and a comparability summary. The results will be presented as patterns in reported 311 demands.

Dataset Biography

A major limitation of the Toronto 311 open dataset is that it does not represent the full universe of 311 activity. According to a Toronto Open Data guest post, the open 311 dataset represents only approximately 30–35% of total 311 requests and covers only 6 of the city's 45 divisions. This means the dataset is suitable for analyzing patterns in the open customer-initiated service request records, but it should not be treated as a complete measure of all 311 activity or all municipal service demand.

The geographic fields also require caution. The dataset includes the first three characters of the postal code, which represents a larger area rather than a specific postal code. This protects privacy by avoiding precise household-level location information, but it also limits the level of geographic analysis that can be performed. The dataset also includes ward information, but direct ward comparison over time is limited because Toronto's ward boundaries changed between the early and recent comparison periods.

These limitations affect how the findings can be interpreted. If one service request category, division, ward, or postal-code area appears to have higher request volume, this should be interpreted as higher reported demand within the open dataset, not as proof of higher actual service need. The dataset's partial coverage limited geographic precision, category changes, division coverage, and ward boundary changes all support the need for data profiling, cautious comparison, and clear limitation statements before drawing conclusions.

Data Structure and Integration Plan

The Toronto 311 data is organized as yearly CSV files, with each row representing one reported service request. The dataset is already in a flat table format, so I do not need to join multiple tables together. The main integration step will be combining the yearly files into one larger dataset. The main columns include Creation Date, Status, First 3 Chars of Postal Code, Ward, Service Request Type, Division, Section, and intersection street fields. Most of these columns are text-based or categorical. The Creation Date column will be converted into a date format so I can create year, month, and season columns for the time-based analysis.

I will load each yearly CSV file in Python using pandas, following the same general approach used in the course labs. Before combining the files, I will check each one using `.head()`, `.shape`, `.info()`, and missing value checks to make sure the files loaded correctly and have consistent columns. After that, I will add columns for year, time period, and source file. The years 2010–2013 will be labelled as the early period, and 2022–2025 will be labelled as the recent period. Some fields will be interpreted carefully. The intersection street columns have many missing values, the postal code field only shows the first three characters, and ward boundaries may have changed over time. Because of this, geographic patterns will be used cautiously and will not be treated as proof of actual neighbourhood need.

EDA / Data Profiling and Audit

For the EDA and data profiling step, I reviewed each yearly Toronto 311 CSV file before combining the data. Since this project compares 2010–2013 with 2022–2025, I needed to check whether the files were similar enough to compare.

I loaded each file in Python using pandas and checked the first few rows, row counts, column names, data types, missing values, duplicate records, and unique values. All files had the same 9 columns: Creation Date, Status, First 3 Chars of Postal Code, Intersection Street 1, Intersection Street 2, Ward, Service Request Type, Division, and Section. This means the files can be combined into one dataset. Even though the column names are the same, the values inside some columns changed over time. For example, the early files have about 44–45 ward values, while the recent files have 26 ward values. Service request types, divisions, and sections also have different numbers of unique values across the two periods. This means some differences may come from changes in classification, not only changes in reported demand. The missing value review showed that the intersection street fields had many missing records, so they may not be useful for the main analysis. I will focus more on fields that support the research questions, such as Creation Date, Service Request Type, Division, Ward, Status, Section, and the first three characters of the postal code.

Overall, the EDA showed that the files can be combined, but some fields need caution before interpretation. The report will include summary tables for row counts, missing values, duplicate checks, and unique values. The full Python notebook will be uploaded to GitHub repository.

Cleaning and Preparation Planning

The cleaning and preparation plan will focus on turning the audited yearly CSV files into one analysis-ready dataset. After the EDA confirms that the files have the same main structure, the 2010, 2011, 2012, 2013, 2022, 2023, 2024, and 2025 files will be combined into one dataset. I will add a source year field and a comparison period field so each record can be identified as part of either 2010–2013 or 2022–2025. The Creation Date field will be converted into a proper date format. From this field, I will create new columns for year, month, month name, weekday, and season. These variables are needed for the monthly and seasonal parts of the analysis.

The main categorical fields will be cleaned and prepared for grouping. These include Service Request Type, Division, Section, Status, Ward, and the first three characters of the postal code. I will remove extra spaces, review formatting issues, and check whether labels changed across years. I will not automatically combine service request categories unless there is a clear reason,

because category changes are part of the project's comparability issue. Missing values will be handled based on the importance of each field. Fields that are central to the research questions, such as Creation Date, Service Request Type, Division, Status, Ward, and Section, will be reviewed carefully. Fields with many missing values, such as the intersection street fields, may be excluded from the main analysis or used only as supporting information. Duplicate records will also be checked before the final analysis because repeated rows could affect request counts, rankings, and percentage shares. After cleaning, the final dataset will keep the fields needed for the research questions and will be used to create the summary tables, charts, seasonal analysis, and comparability audit.

Exploratory Data Analysis Results

The exploratory data analysis was completed using the combined Toronto 311 dataset for the selected early and recent comparison periods. The final combined dataset included 2,913,502 records from 2010–2013 and 2022–2025. The early period contained 1,232,720 records, while the recent period contained 1,680,782 records.

The data quality review showed that the main fields used in the analysis were complete. Creation Date, Status, Ward, Service Request Type, Division, Section, Source Year, and Period had no missing values. The main missingness issue was found in the intersection street fields. Intersection Street 1 was missing 87.50% of its values, and Intersection Street 2 was missing 87.56%. Because of this, intersection-level analysis was not used as a main part of the project. A duplicate check found 2,318 duplicate rows, which represented about 0.08% of the full dataset. Since the dataset does not include a unique service request ID, these duplicates were documented as a data quality issue rather than removed automatically.

The service request type analysis showed that reported 311 demand is spread across many categories. The dataset contained 970 unique service request types. The largest overall category was Property Standards, but it represented only 3.24% of all records. The Pareto-style analysis showed that the top 20 service request types accounted for only 34.38% of all records. This means the data is not dominated by one or two request types. Instead, reported demand is distributed across a wide range of categories.

The comparison between the early and recent periods showed changes in the highest-volume service request types. In the early period, Residential Furniture / Not Picked Up, Property Standards, and Sewer Service Line-Blocked were among the highest-volume categories. In the recent period, Injured - Wildlife became the highest-volume category, followed by Residential: Bin: Repair or Replace Lid and Property Standards. This suggests that the composition of reported 311 demand changed across the two time periods.

The division-level analysis also showed changes over time. Solid Waste Management Services remained the largest division in both periods, but its share decreased from 44.64% in the early period to 36.22% in the recent period. Municipal Licensing & Standards increased from 13.12% to 26.21%, and Transportation Services increased from 15.98% to 25.40%. Toronto Water decreased from 15.76% in the early period to 8.22% in the recent period. These results suggest that the distribution of reported 311 demand across City divisions changed between the two periods.

The monthly and seasonal analysis showed that 311 request volume is generally higher in warmer months, but the pattern varies by request type. Summer had the highest overall share in both periods, but the recent period had a more balanced seasonal distribution than the early period. The heatmap of the top 10 service request types showed that different categories follow different monthly patterns. For example, Road - Pot hole had higher activity around March and April, while Property Standards and Injured - Wildlife were higher in warmer months. This supports the second research question by showing that monthly patterns are not the same across all request categories.

Baseline Model and Evaluation

For Milestone 3, I added a simple baseline model to test whether recent monthly request patterns could provide a basic prediction of 2025 monthly request volume. This model is connected to RQ2 because it focuses on monthly request patterns and seasonal reporting behaviour.

The baseline model used 2022, 2023, and 2024 as the training period and 2025 as the testing period. For each month in 2025, the predicted value was calculated by taking the average request count for that same month across 2022, 2023, and 2024. For example, the January 2025 prediction was based on the average of January 2022, January 2023, and January 2024. This setup avoids leakage because the actual 2025 values were not used to create the predictions. The 2025 data was only used after the predictions were created, so the actual and predicted values could be compared.

The baseline model produced a Mean Absolute Error (MAE) of 5,578.75, a Root Mean Squared Error (RMSE) of 8,201.21, and a Mean Absolute Percentage Error (MAPE) of 13.39%. The MAE means the model was off by about 5,579 requests per month on average. The RMSE was higher than the MAE because some months had larger errors, especially February and December 2025. The MAPE shows that the baseline predictions were off by about 13.39% on average.

The actual versus predicted chart showed that the baseline model followed the general monthly pattern in some parts of the year, especially from May to November. However, it did not capture larger spikes in February and December. This means the model is useful as a simple comparison point, but it should not be treated as a final predictive model. The results show that monthly 311 request patterns have some stability, but there is still uncertainty in future request volume.

Evaluation Criteria, Assumptions, and Uncertainty

The baseline model was evaluated using MAE, RMSE, and MAPE because these metrics show how far the predicted monthly counts were from the actual 2025 monthly counts. Lower error values would indicate stronger performance. The main assumption is that monthly patterns from 2022–2024 provide a reasonable comparison point for 2025. However, this assumption is limited because 311 request volume can be affected by reporting behaviour, category changes, service changes, public awareness, weather events, and other factors not included in the model. This setup avoids leakage because the actual 2025 values were not used to create the predictions.

The model does not explain why request volume changed. It only tests whether recent monthly averages can provide a basic expected value for 2025. Because of this, the results should be interpreted as a baseline comparison, not as a final forecasting tool or causal model.

Pipeline Diagram and Sequence of Operations

The pipeline for this project shows the order of steps from the raw yearly CSV files to the final analysis outputs. The purpose of the pipeline is to make the project easier to follow and easier to reproduce.

Figure 1.1

Pipeline Diagram and Sequence of Operations

Milestone 3 workflow from raw data inputs to EDA, baseline modeling, evaluation, and final outputs.

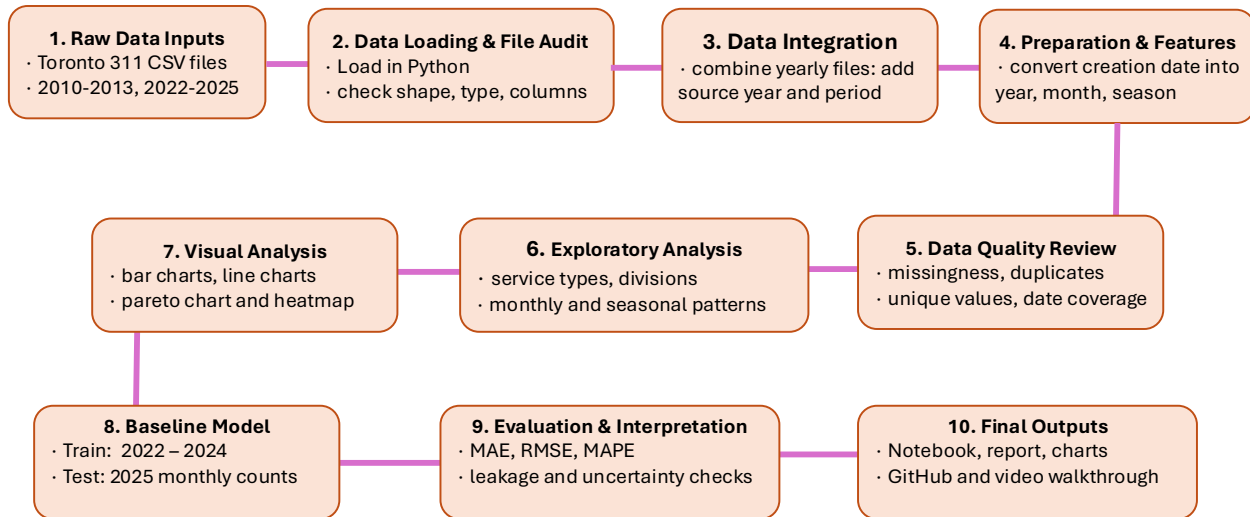


Figure 1.1 shows the project workflow from the raw Toronto 311 yearly CSV files to the final project outputs. The process begins with data loading and file auditing, followed by data integration, preparation, and feature engineering. After data quality review, the project moves into exploratory analysis, visual analysis, and a simple baseline monthly prediction model. The final stages include model evaluation, interpretation of limitations, and final outputs such as the notebook, report, charts, GitHub repository, and video walkthrough.

Limitations: Association, Not Causation

This project can show patterns in Toronto 311 service request data, but it cannot prove what caused those patterns. If one request type increased between 2010–2013 and 2022–2025, the data can show that the increase appears in the available records, but it cannot prove that actual municipal need increased by the same amount.

A major limitation is that 311 data reflects ‘customer’ (emphasis on customer because it’s consumer based and so it comes with bias on what they deem as a concern or what they feel they should be requesting) reported demand, not all community need. A higher number of requests may mean there are more service issues, but it may also reflect higher awareness of 311, easier access to reporting tools, repeated reports, or greater willingness to contact the City. A lower number of requests also does not automatically mean fewer problems, because some residents may face barriers such as language, time, technology access, or lower trust in government.

The articles I reviewed also point out that 311 data can be hard to interpret. White and Trump warn that 311 data usually does not show who made the request, and a small number of frequent users may affect request totals. Wang et al. and Stowers also show that service request categories can change over time. This matters for my project because service request labels, division names, section labels, and ward values may not be fully comparable between 2010–2013 and 2022–2025.

There are also limits with the Toronto dataset itself. The open 311 dataset only represents part of total 311 activity, and some fields have missing values or limited detail. The data can show that a request was made, but it does not fully explain the severity of the issue, the root cause, or whether the resident was satisfied with the result. These points are to be kept in mind when working with the data and not seeing the results as an absolute truth or the answer to forecasting requests. There are limitations and possible bias!

Privacy, Misuse, Bias, and AI Use

This project uses data from the City of Toronto's 311 Service Requests dataset. The dataset does not include and personal or identifying information like names, phone numbers, email addresses, or full addresses. Geographic information is limited to broader fields such as ward, intersections, and the first three characters of the postal code, which reduces the risk of identifying individual residents or households. The main privacy risk is not direct identification, but over-interpreting local service request patterns. Results will be reported at a broader level, such as service request type, division, period, month, season, or ward. Bias is also a key limitation because 311 data only show reported requests. A high number of requests does not automatically mean greater actual need, and a low number of requests does not mean fewer problems. Request volume may reflect awareness of 311, access to reporting channels, language barriers, trust in government, or repeated reports. The open dataset also represents only part of total 311 activity, so findings will be described as patterns in reported demand, not proof of actual municipal need.

I used ChatGPT to help explain some coding concepts I was not as familiar with, fix grammatical errors within the paragraphs written, and fix some of my outputs visually. ChatGPT helped me organize differently designed tables, colours on bar graphs and the heatmap. A lot of the coding is from previous knowledge and coursework taken during this program. Things beyond my knowledge I went to Python forums to learn and understand and used ChatGPT to help me further understand and learn how the code is written amongst other codes. I used ChatGPT to help explain coding concepts, improve wording, and troubleshoot visual outputs such as tables, bar charts, and the heatmap. The coding decisions, dataset interpretation, results, and final analysis remain my own. Many of the coding steps are based on previous coursework and lab-style Python work from the program. When I used AI assistance, I checked the suggestions against my dataset, course instructions, uploaded articles, and City of Toronto sources. I understand that AI can produce incorrect code, fake citations, or unsupported interpretations, so I reviewed the outputs before including them in the project.

Github Repository and Reproducibility

A GitHub repository was created to organize the project files and support reproducibility. The repository includes the Milestone 3 Python notebook, README file, written report, and supporting project outputs. The notebook documents the main analysis steps, including loading

the yearly CSV files, combining the selected years, creating the Source Year and Period fields, converting Creation Date into datetime format, creating time-based fields, completing data quality checks, producing EDA tables and charts, and running the baseline monthly prediction model.

For this project, reproducibility means that another person should be able to understand what data was used, what steps were followed, and how the results were produced. The raw data comes from the City of Toronto Open Data Portal. To rerun the notebook, the same yearly CSV files for 2010, 2011, 2012, 2013, 2022, 2023, 2024, and 2025 must be downloaded and placed in the correct project folder. The notebook can then be rerun from top to bottom to reproduce the combined dataset, summary tables, charts, Pareto analysis, heatmap, and baseline model results.

The repository supports transparency by showing the code, analysis workflow, assumptions, limitations, and final outputs in one location.

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